

Master class: Wireshark 101

Introduction to Network packet Analysis and packet filtering

What We'll Cover

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| 2 Understanding Network Packet Capture | 7 Using Filter Operators and |
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Introduction to Wireshark

What is Wireshark?

- ✓ A free, open-source network protocol analyzer
- ✓ Captures and displays data traveling on a network in real time

Role as a Packet Analyzer / Traffic Sniffer

- ✓ Inspects individual packets at a granular level
- ✓ Helps diagnose network issues, security incidents, and protocol behavior

Widely used in security, IT, and forensics

- ✓ Standard tool for network troubleshooting and penetration testing



Understanding Network Packet Capture

Selecting the Network Interface

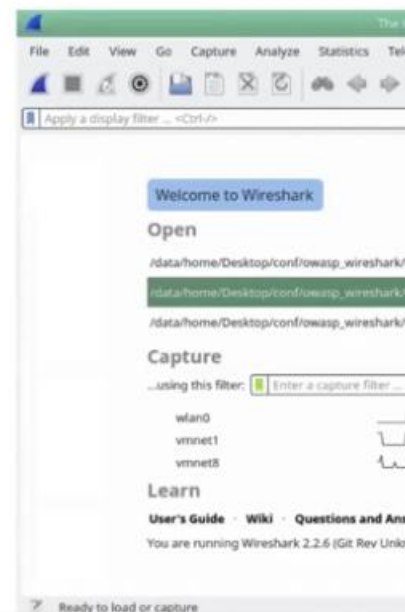
- ✓ Choose the correct adapter (Wi-Fi, Ethernet, VPN, etc.)
- ✓ Wireshark lists live traffic activity per interface before capture

Starting a Packet Capture

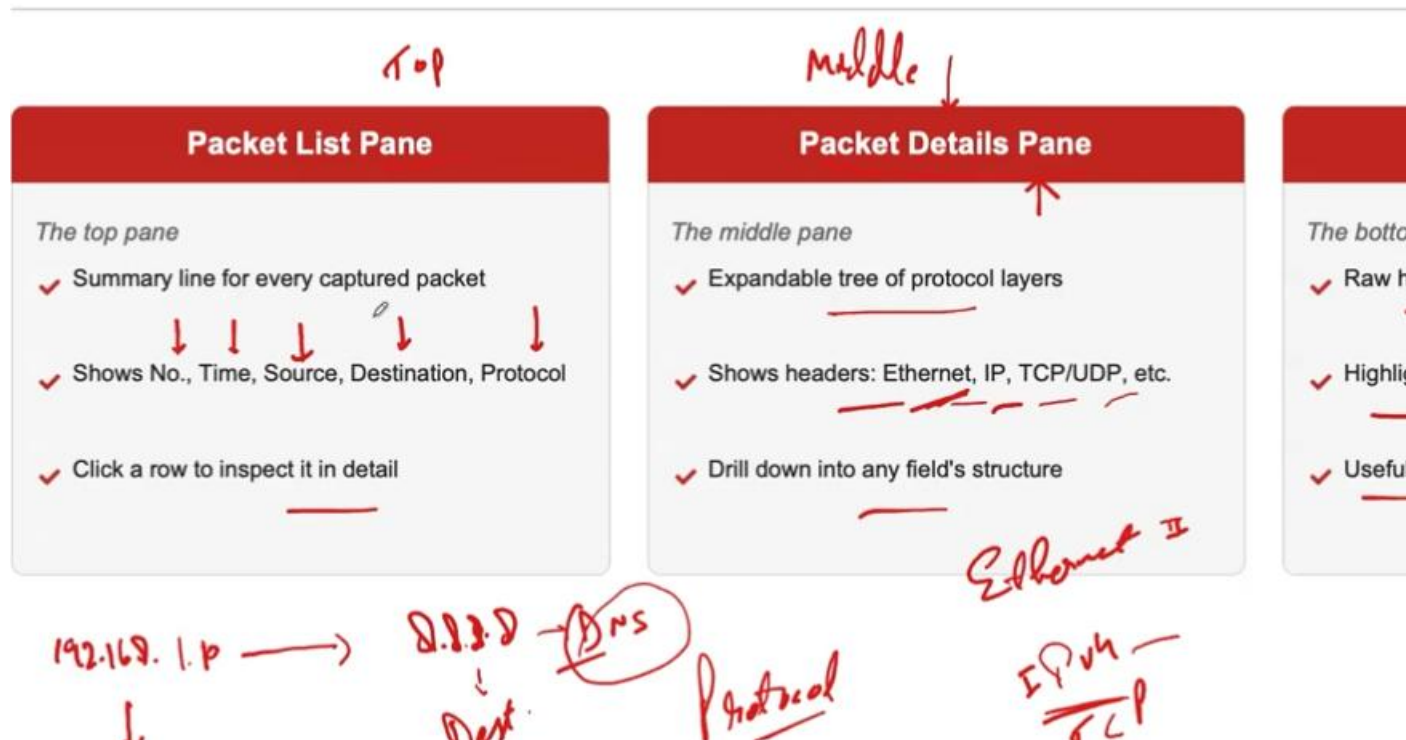
- ✓ Click the shark-fin icon or press Ctrl+E to begin capturing
- ✓ Packets appear live in the Packet List pane as they arrive

Stopping a Packet Capture

- ✓ Click the red square icon or press Ctrl+E again to stop



Understanding the Wireshark Interface



Reading Packet Information

Source / Destination

The sending and receiving IP addresses of a packet

Protocol

The protocol used — e.g. TCP, UDP, DNS, HTTP, HTTPS

Packet Length

Total size of the packet in bytes

Ports

Source and destination port numbers identifying the service

No.	Time	Source	Destination
13	1.256128834	192.168.178.27	192.168.178.27
14	1.272047640	193.70.91.56	192.168.178.27
15	1.503106801	192.168.178.27	192.168.178.27
16	2.992854539	192.168.178.27	192.168.178.27
17	3.732746500	193.70.91.56	192.168.178.27
18	3.732785793	193.70.91.56	192.168.178.27
19	3.732809890	192.168.178.27	192.168.178.27

> Frame 10: 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface 0
> Linux cooked capture
> Internet Protocol Version 4, Src: 192.168.178.27, Dst: 192.168.178.27
> Transmission Control Protocol, Src Port: 5555, Dst Port: 80, Seq: 304412544, Win: 65535, Len: 0

0000	00 04 00 01 00 06 b8 08	cf 58 61 64 00
0010	45 00 00 3c c9 c5 40 00	40 06 e1 b3 c0
0020	c1 46 5b 38 09 26 00 50	e6 3e 0e f3 00
0030	a0 02 72 10 41 2c 00 00	02 04 05 b4 04
0040	61 f2 d4 e5 00 00 00 00	01 03 03 07

Introduction to Wireshark Filters

Why Use Display Filters?

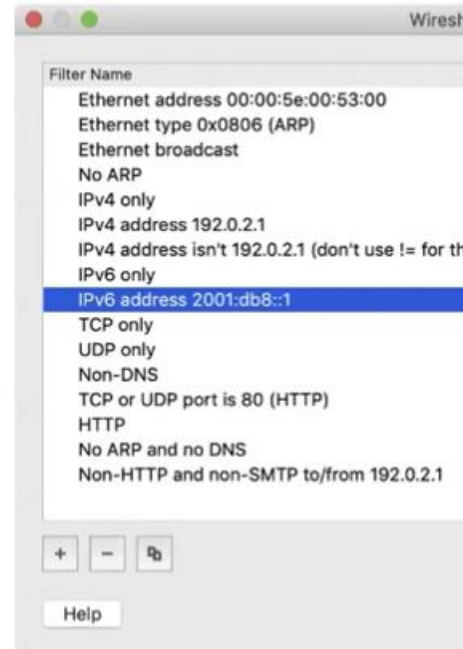
- ✓ Raw captures can contain thousands of packets
- ✓ Filters narrow the view to only relevant traffic

Where Filters Are Applied

- ✓ Typed into the display filter bar at the top of Wireshark
- ✓ Green bar = valid syntax; red bar = invalid syntax

Result

- ✓ Only packets matching the filter expression remain visible



Common Display Filters

tcp / udp

Show only TCP or UDP traffic

dns

Show only DNS query and response packets

http / https

Show web traffic over HTTP or HTTPS (TLS)

ip.addr / tcp.port

Filter by a specific IP address or port number

Using Filter Operators and Expressions

== / !=

Equal to / not equal to a given value

> / <

Greater than / less than a given value

contains

Field or payload contains a given substring

and / or / not

Combine or negate multiple filter expressions

Using Filter Operators and Expressions

== / !=

Equal to / not equal to a given value

ip.addr == 192.168.10.1
ip.addr != 192.168.10.1

> / <

Greater than / less than a given value

tcp.len > 1000
frame.len < 100

contains

Field or payload contains a given substring

http.request.uri contains
tcp contains

and / or / not

Combine or negate multiple filter expressions

Using Filter Operators and Expressions

== / !=

Equal to / not equal to a given value

ip.addr == 192.168.10.1
ip.addr != 192.168.10.1

> / <

Greater than / less than a given value

tcp.len > 1000
frame.len < 100

contains

Field or payload contains a given substring

http.request.uri co
tcp con

and / or / not

Combine or negate multiple filter expressions

TRUE

ip.addr == 192.168.10.1 and tcp.port == 443

Analyzing Web Traffic

Examining HTTP Requests & Responses

- ✓ Inspect request methods: GET, POST, PUT, DELETE
- ✓ Check response codes: 200 OK, 301, 404, 500

Key Fields to Review

- ✓ Host, URI, headers, and packet content payload
- ✓ Use filter: http.request or http.response

No.	Time	Source	Destination
22	5.110731826	IntelICor_58:61:64	
23	5.362390056	127.0.0.1	127.0.0.1
24	5.362390184	127.0.0.1	127.0.0.1
25	5.362405598	127.0.0.1	127.0.0.1
26	5.362506833	127.0.0.1	127.0.0.1
27	5.362518970	127.0.0.1	127.0.0.1
28	5.362599242	127.0.0.1	127.0.0.1
29	5.362606016	127.0.0.1	127.0.0.1
30	5.362841637	127.0.0.1	127.0.0.1
31	5.363809929	127.0.0.1	127.0.0.1
32	5.363964990	127.0.0.1	127.0.0.1
33	5.364140080	127.0.0.1	127.0.0.1



Following Network Conversations

🛡️ What Is 'Follow Stream'?

- ✓ Reassembles all packets in one TCP/UDP conversation
- ✓ Right-click a packet → Follow → TCP/UDP/HTTP Stream

🛡️ Why It's Useful

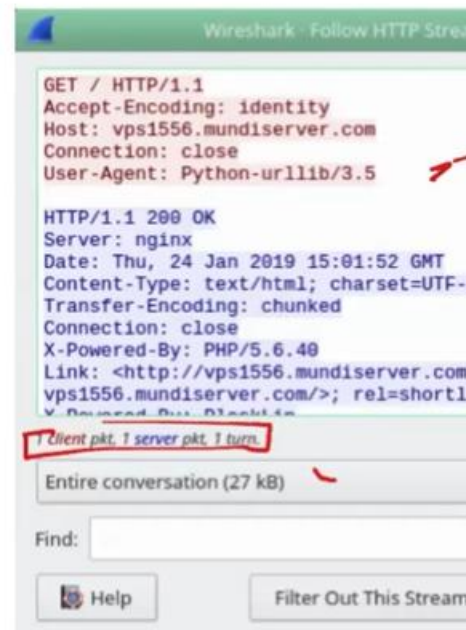
- ✓ View a full exchange between client and server in order
- ✓ Colors distinguish each side of the conversation

🛡️ Common Use Cases

- ✓ Reconstructing credentials, files, or full HTTP sessions

Red color-> client response

Blue color -> server response



Searching and Sorting Captured Traffic

Sort by Column

Click any column header to sort packets by that field

Find Packet

Ctrl+F opens search across packet contents, bytes, or fields

Search Scope

Search within packet list, details, or bytes pane

Search Direction

Search forward or backward from the current packet

Searching and Sorting Captured Traffic

Sort by Column

Click any column header to sort packets by that field

Find Packet

Ctrl+F opens search across packet contents, bytes, or fields

Search Scope

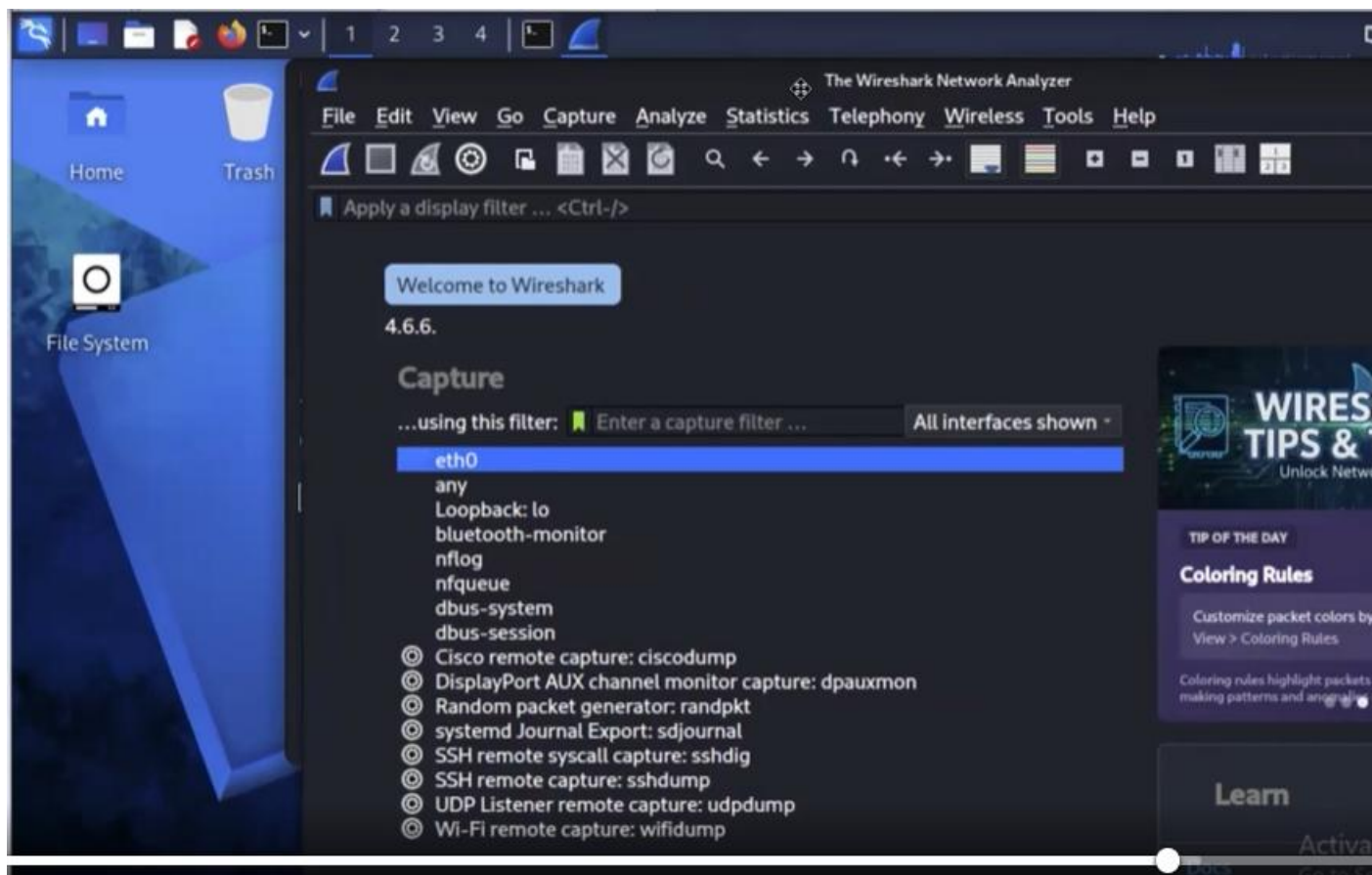
Search within packet list, details, or bytes pane

Search Direction

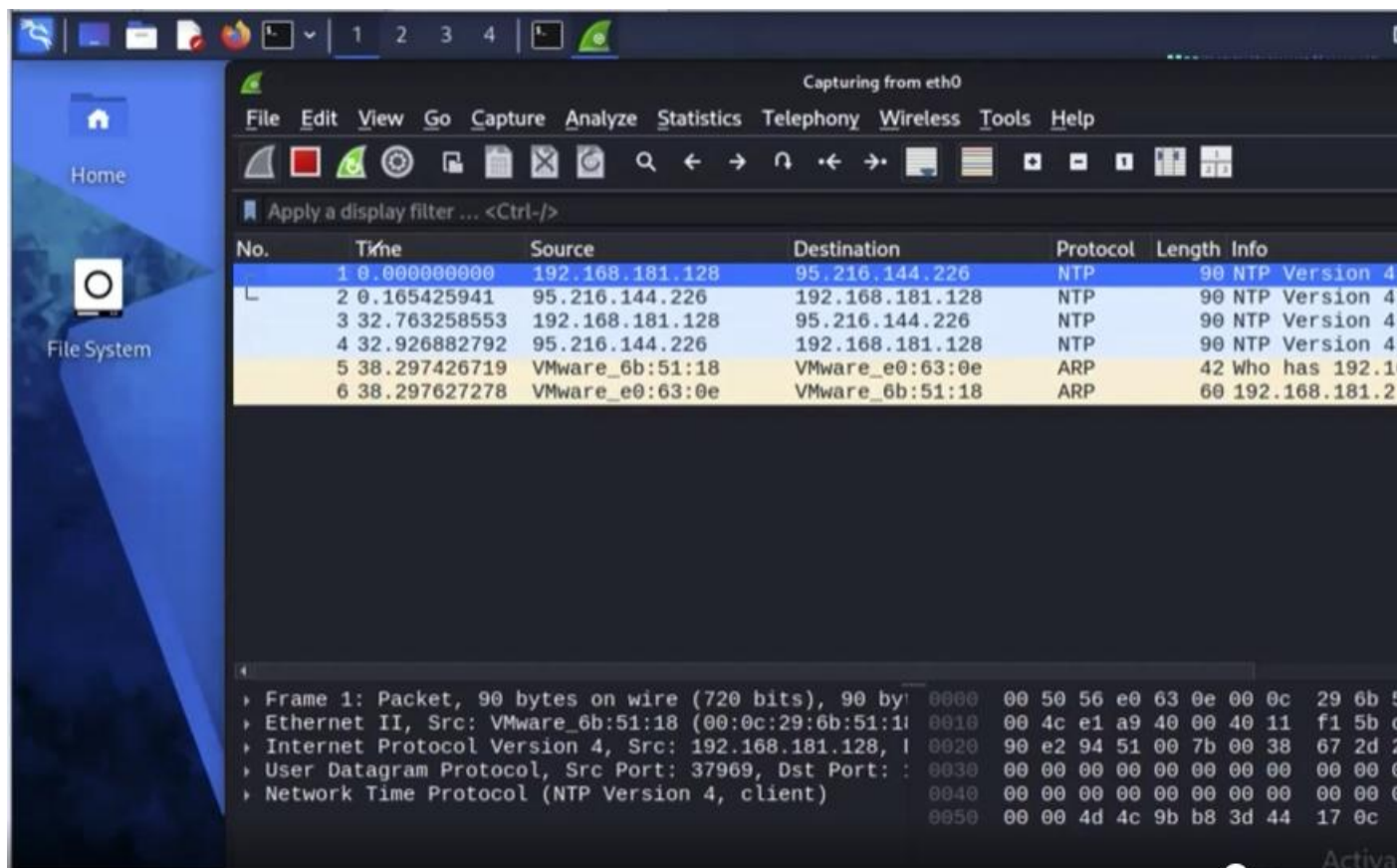
Search forward or backward from the current packet

Open Kali-linux terminal

Type wirshark-> choose eth0

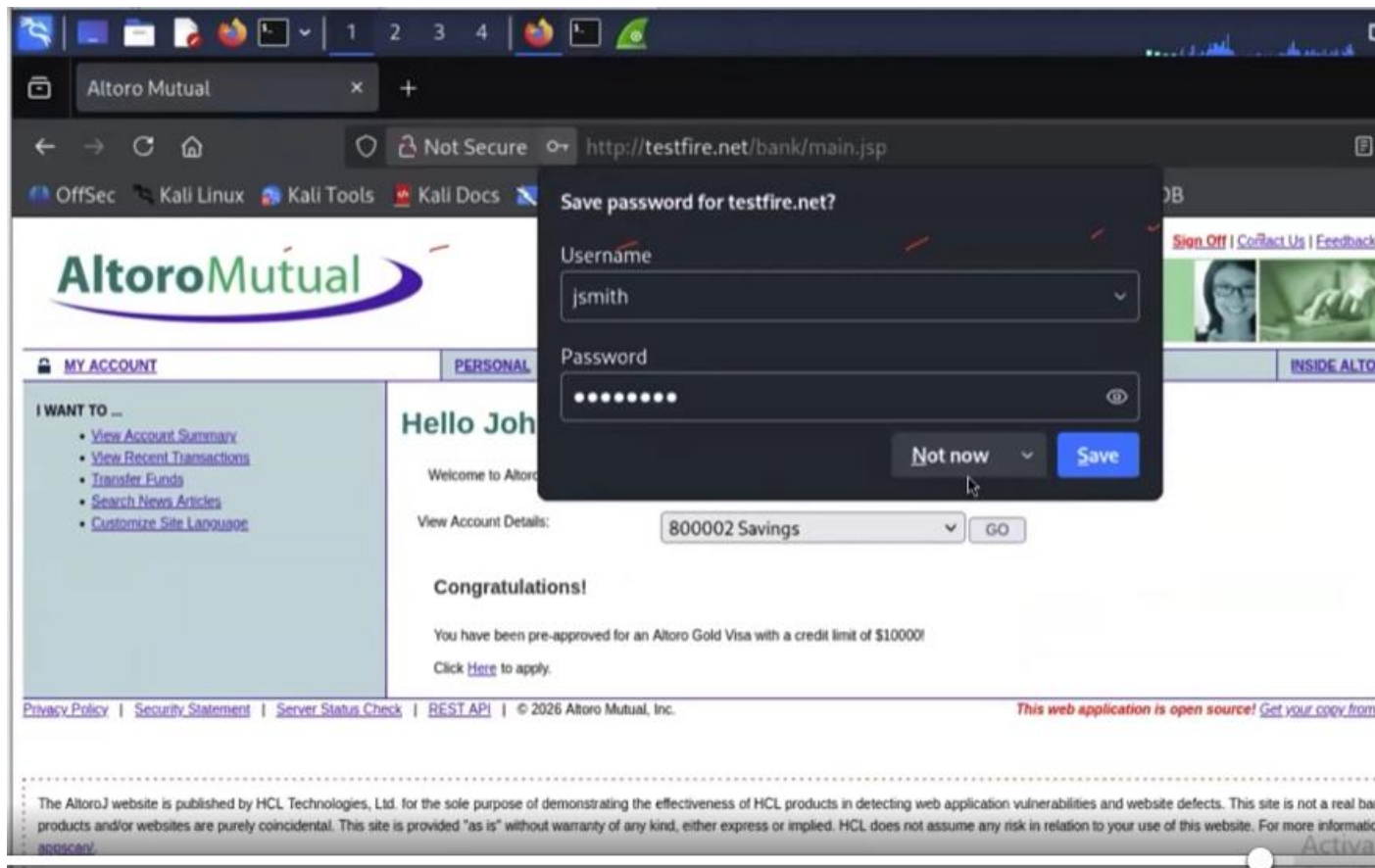


PACKET Summary pane:



Open a browser and hit <http://testfire.net> , Open the login page and then Type username and password

Wireshark experiment: Open <http://testfire.net>



Home x Metasploitable2-Linux x kali-linux-2026.2-vmware... x

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File Edit View Go Capture Analyze Statistics Telephony Wireshark

http

No.	Time	Source	Destination
239	2.952013087	192.168.181.128	65.61.137.1
241	2.952315323	192.168.181.128	65.61.137.1
243	3.198860695	65.61.137.117	192.168.181.128
247	3.201307971	192.168.181.128	65.61.137.1
248	3.201414128	65.61.137.117	192.168.181.128
253	3.204689393	65.61.137.117	192.168.181.128
259	3.208046210	65.61.137.117	192.168.181.128
267	3.452491034	65.61.137.117	192.168.181.128
268	3.462960050	65.61.137.117	192.168.181.128
271	3.518489915	192.168.181.128	65.61.137.1
273	3.775423079	65.61.137.117	192.168.181.128
322	14.880606025	192.168.181.128	65.61.137.1
336	15.137356263	65.61.137.117	192.168.181.128
416	25.139301155	192.168.181.128	65.61.137.1
418	25.396712418	65.61.137.117	192.168.181.128
420	25.405254330	192.168.181.128	65.61.137.1
422	25.663459808	65.61.137.117	192.168.181.128

Frame 416: Packet, 626 bytes on wire (5008 bits), 6: Ethernet II, Src: VMware_6b:51:18 (00:0c:29:6b:51:18), Dst: 65.61.137.117, Protocol: Internet Protocol Version 4, Src: 192.168.181.128, Dst: 65.61.137.117, Length: 60, encapsulation: Transmission Control Protocol, Src Port: 55554, Dst Port: 80, encapsulation: Hypertext Transfer Protocol

HTML Form URL Encoded: application/x-www-form-urlencoded

- Form item: "uid" = "jsmith"
- Form item: "passw" = "Demo1234"
- Form item: "btnSubmit" = "Login"

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- Deposit Product
- Checking
- Loan Products
- Cards
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- Other Services

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Kali-linux-ip as src 192.168.181.128 and dst IP is http://testfire.net

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Wireshark

File Edit View Go Capture Analyze Statistics Telephony Wire

ip.src==192.168.181.128

No.	Time	Source	Destination
394	15.988686484	192.168.181.128	65.61.137.1
397	15.988969160	192.168.181.128	65.61.137.1
400	16.254218014	192.168.181.128	65.61.137.1
402	22.622650617	192.168.181.128	65.61.137.1
405	22.880771842	192.168.181.128	65.61.137.1
409	22.883140959	192.168.181.128	65.61.137.1
410	22.883456941	192.168.181.128	65.61.137.1
414	23.126472295	192.168.181.128	65.61.137.1
416	25.139301155	192.168.181.128	65.61.137.1
419	25.396754608	192.168.181.128	65.61.137.1
420	25.405254330	192.168.181.128	65.61.137.1
423	25.663795178	192.168.181.128	65.61.137.1
424	29.477194300	192.168.181.128	65.61.137.1
426	29.730649434	192.168.181.128	65.61.137.1
427	29.732534277	192.168.181.128	65.61.137.1
430	29.755308818	192.168.181.128	65.61.137.1
431	29.755803544	192.168.181.128	65.61.137.1

Frame 416: Packet, 626 bytes on wire (5008 bits), 6: Ethernet II, Src: VMware_6b:51:18 (00:0c:29:6b:51:18), Dst: 65.61.137.1, Internet Protocol Version 4, Src: 192.168.181.128, Dst: 65.61.137.1, Transmission Control Protocol, Src Port: 55554, Dst Port: 80, Hypertext Transfer Protocol

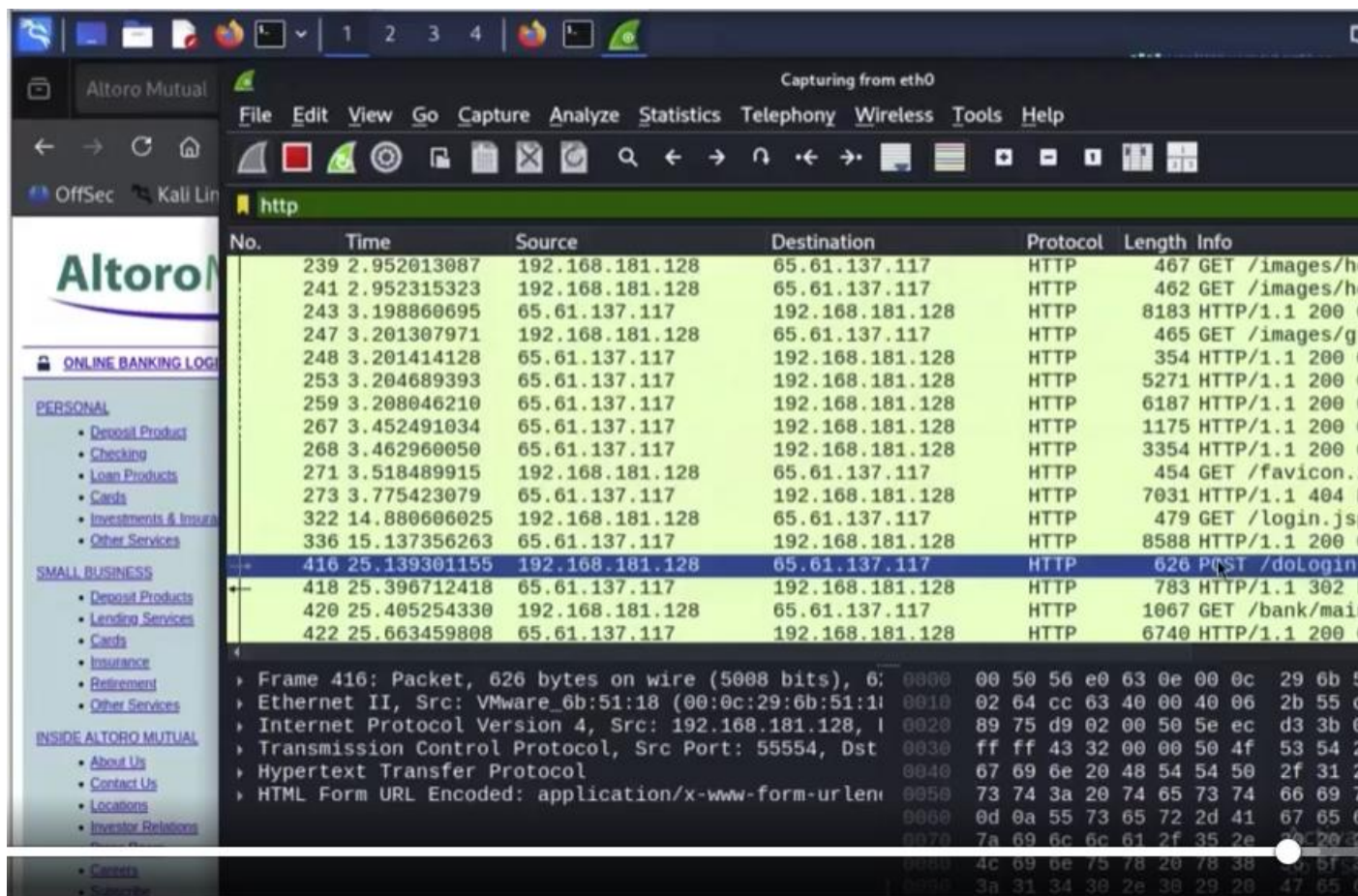
HTML Form URL Encoded: application/x-www-form-urlencoded

- Form item: "uid" = "jsmith"
- Form item: "passw" = "Demo1234"
- Form item: "btnSubmit" = "Login"

Protocol: http -> follow stream

Follow graph option

Right click the session and choose follow http stream option



Home x Metasploitable2-Linux x kali-linux-2026.2-vmware... x

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Wireshark - Follow HTTP Stream (tcp.strea

File

GET / HTTP/1.1
Host: testfire.net
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:40.0
Accept: text/html,application/xhtml+xml,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Upgrade-Insecure-Requests: 1
Priority: u=0, i

No.

HTTP/1.1 200 OK
Server: Apache-Coyote/1.1
Set-Cookie: JSESSIONID=C234F73E80F427B673231CCF
Content-Type: text/html; charset=ISO-8859-1
Transfer-Encoding: chunked
Date: Fri, 21 Aug 2026 09:12:47 GMT

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Int
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Hyp

<!-- BEGIN HEADER -->
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

